Predicting Coronary Heart Disease from BRFSS Survey Data with Linear, Logistic and Tree Models

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***Abstract*—We predict self-reported coronary heart disease from the 2015 BRFSS survey, 8.8% positive, with linear, logistic and tree models in NumPy, choosing everything, threshold included, by nested cross-validation of F1. Regularized logistic regression on codebook-cleaned features with one-hot encodings, missing indicators and interactions reaches F1 0.441 ± 0.006, 0.450 on AIcrowd and 0.454 blended with boosted trees. The codebook rules help most; regularization and class weighting do not.**

I. Introduction

Cardiovascular disease is the leading cause of death worldwide. To predict self-reported coronary heart disease or myocardial infarction (MICHD) from the 2015 BRFSS survey, we contribute (a) a label-free preprocessing built from the BRFSS codebook (Section II); (b) a nested cross-validation protocol in which the threshold is also chosen without seeing the test fold (Section III-A); (c) a comparison of linear and logistic models and of class weighting, and a final model with interactions, blended with boosted trees (Sections III–III-D); and (d) ablations showing which choices actually improve F1 (Section IV).

II. Data and preprocessing

The 328,135 training and 109,379 test respondents are described by 321 features, mostly coded answers; 8.8% of the training respondents are positive. 44.8% of the values are missing, and 147 features, mostly from optional state modules, for more than half of the respondents.

The features are coded survey answers, so we first convert them into meaningful values with rules from the BRFSS 2015 codebook [1], checked for every feature (Table I).

Missing values are then set to the training median, constant features are dropped, the others standardized and a bias added, with statistics from the training rows only; alone, this is the generic baseline (Table III).

Rule	Example
Don't know, refused → missing	codes 7 and 9
None → 0	88 = "no days", 888, 555
Drop 32 survey features	identifiers, weights
Drop 29 raw answers	WEIGHT2, cleaned in WTKG3
Drop 3 nominal features	activity types, kept in computed minutes; one 99.9% missing

Table I
Codebook rules, applied before any learning.

The final model also adds binary features: a one-hot encoding of the nominal features and of every feature with 3 to 10 values, and a missingness indicator for each feature with missing values. The indicators are informative because many questions are asked only after a given answer: the age at diabetes diagnosis is missing for respondents without diabetes.

III. Methods

A. Evaluation protocol

Models are evaluated by nested cross-validation over five stratified folds shared by all experiments. For each test fold, the four other folds estimate the preprocessing statistics and, by an inner 3-fold cross-validation, select the hyperparameters and the threshold t that maximize F1; the selected model is retrained on them and evaluated on the test fold, which thus never influences a choice. We report the mean and standard deviation of F1 over the test folds. Two configurations are compared by their per-fold F1 difference Δ, and one is better only if $\Delta > 0$ in all five folds. The final model results from the same selection on the full training set.

B. Linear models

The linear models minimize $\frac{1}{2N}\|y - Xw\|^2$, plus $\lambda\|w\|^2$ for ridge, $\lambda \in \{10^{-6}, \dots, 10^{-1}\}$, and threshold $x_i^\top w$. Least squares and ridge are solved exactly; from $w = 0$, GD takes 200 steps, $\gamma \in \{1, 3, 10, 30, 60, 90\} \cdot 10^{-3}$, and SGD 200,000 single-sample steps, about one epoch, $\gamma \in \{1, 3\} \cdot 10^{-k}$, $k = 4, 5, 6$, seed 0.

C. Logistic models

Logistic regression models the probability of disease as $\sigma(x_i^\top w)$ and minimizes the mean negative log-likelihood $L(w) = \frac{1}{N}\sum_i \left[\log(1+e^{x_i^\top w}) - y_i x_i^\top w\right]$; its regularized version minimizes $L(w) + \lambda\|w\|^2$. Both use full-batch gradient descent with step size γ from $w = 0$ for 300 iterations, after which the loss changes by less than 0.06%. The inner cross-validation selects $\gamma \in \{0.01, 0.03, 0.1, 0.3, 1\}$, $\lambda \in \{10^{-6}, \dots, 10^{-1}\}$ and $t \in \{0.05, 0.06, \dots, 0.95\}$.

D. Final model

A linear model cannot capture joint effects, such as smoking being more dangerous for older respondents. The final model therefore fits regularized logistic regression, ranks the standardized features by $|w_j|$, and refits it with the $K(K-1)/2$ standardized products of the top K, with $K \in \{20, 30, \ldots, 80\}$ chosen by 3-fold cross-validation. The submitted predictions blend the final model with gradient-boosted trees, implemented in NumPy: 200 trees of depth 4, each a Newton step of size 0.08 on the logistic loss, on the cleaned features cut into 32 quantile bins plus one for missing values. The blend predicts disease when $0.8\,p_{\text{trees}} + 0.2\,p_{\text{final}} \geq 0.222$, the weight and threshold that maximize inner-validation F1 without lowering the final model's accuracy.

IV. Results

The final model is the best method (Table II), with F1 $= 0.441 \pm 0.006$ in nested cross-validation and 0.450 on AIcrowd. Its gain does not come from the hyperparameters: F1 is flat around the selected $\gamma = 0.3$, $\lambda = 10^{-4}$ and $K = 40$ (Fig. 1), and every outer fold selects $\gamma \in \{0.3, 1\}$ and $\lambda \leq 10^{-3}$. Nor does it come from regularization: plain and regularized logistic regression are tied, $\Delta = 0.0003 \pm 0.0014$, as training and validation losses of 0.219 and 0.221 show no overfitting. The linear models are weaker: regularized logistic regression beats least squares in all five folds, $\Delta = 0.011 \pm 0.002$, mostly through precision, 0.358 against 0.346, as the squared loss also penalizes correct predictions beyond 0 or 1.



Figure 1. Inner-CV F1 on the full training set; $\lambda = 0$: plain logistic regression; orange: selected.

The gain therefore comes from the features, added one at a time in Table III. The codebook rules contribute most, raising precision from 0.337 to 0.361. One-hot encoding and missing indicators help in all folds, the interactions in four folds only, a likely but uncertain gain. Squared or binned features, more interactions and model averaging changed F1 by less than 0.001.

Finally, the class imbalance needs no correction beyond the threshold. Weighting the loss of the positive samples by $\omega = 10.3$, the ratio of negatives to positives, or by its square root 3.2 gives F1 0.430 and 0.431 against 0.432, but thresholds 0.75 and 0.48 instead of 0.21. Weighting multiplies the odds by about ω, so only the threshold moves: for $\omega = 10.3$, 0.21 becomes 0.73.

On the same outer folds, with weight and threshold chosen on a held-out quarter of each training part, the tree blend beats the final model in four folds, F1 0.444 ± 0.007 against 0.441 ± 0.007, at equal accuracy, 0.885 against 0.884, and reaches 0.454 on AIcrowd. Deeper or slower trees did not help.

Method	F1	Prec.	Rec.	Acc.	Thr.
GD (MSE)	0.419 ± 0.005	0.344	0.536	0.869	0.22
SGD (MSE)	0.415 ± 0.005	0.337	0.544	0.865	0.20
Least squares	0.420 ± 0.005	0.346	0.537	0.869	0.22
Ridge	0.420 ± 0.004	0.343	0.542	0.868	0.22
Logistic	0.431 ± 0.005	0.359	0.540	0.874	0.20
Reg. logistic	0.432 ± 0.005	0.358	0.543	0.874	0.20
Final model	$\mathbf{0.441 \pm 0.006}$	0.369	0.549	0.877	0.21

Table II
Nested cross-validation, mean over the 5 test folds; Thr.: mean selected threshold. Rows above the last one-hot encode nominal features only. Always predicting "no disease": accuracy 0.912, F1 = 0.

Configuration	F1	Δ
Generic baseline	0.416 ± 0.005	
+ codebook rules	0.429 ± 0.003	$+0.013 \pm 0.004$
+ one-hot encoding	0.433 ± 0.005	$+0.004 \pm 0.003$
+ missing indicators	0.438 ± 0.004	$+0.005 \pm 0.002$
+ 40-feature interactions	0.441 ± 0.006	$+0.003 \pm 0.002$

Table III
From the baseline to the final model, one change per row; Δ: per-fold gain over the row above.

V. Conclusion

Codebook preprocessing matters most, and blending the final model with gradient-boosted trees raises the AIcrowd F1 from 0.450 to 0.454. The labels are self-reported and the survey is observational, so the models capture associations, not causes, and with a precision below 0.4 most predicted positives are false alarms. The small, uncertain gains of the interactions and of the trees suggest that these features are close to their limit.

Reproducibility

Code: **[public GitHub URL and commit]**. `run.py` with `config/boosted_v2.json` reproduces the submitted predictions; the README lists the commands behind every table and figure.

REFERENCES

[1] Centers for Disease Control and Prevention, "Behavioral risk factor surveillance system: LLCP 2015 codebook report," https://www.cdc.gov/brfss/annual_data/2015/pdf/codebook15_llcp.pdf, 2016, accessed October 2026.